Today

- More on number state vector space of SHO
- The Schrödinger equation in three dimensions
- Angular momentum
- Orbital angular momentum and spherical harmonics
- The hydrogen atom

3.8.2 Normalized number state vectors

Because the energy levels of the simple harmonic oscillator in one dimension are not degenerate, the states $|n-1\rangle$ and $\hat{a}|n\rangle$ must be physically the same, and could only differ by a constant factor. Thus, we write

$$|n-1\rangle = c_n \hat{a}|n\rangle, \tag{3.8.12}$$

where c_n is a constant. The constant c_n follows from the normalization of the states. In particular, considering the matrix element of $\hat{N}$,

$$\langle n|\hat{N}|n\rangle = \langle n|\hat{a}^\dagger \hat{a}|n\rangle = |\hat{a}|n\rangle|^2 . \tag{3.8.13}$$

On the other hand,

$$\langle n|\hat{N}|n\rangle = n\langle n|n\rangle = n \tag{3.8.14}$$

for normalized $|n\rangle$. Therefore,

$$1 = \langle n-1|n-1\rangle = |c_n|^2 \langle n|\hat{N}|n\rangle = |c_n|^2 n, \tag{3.8.15}$$

or, choosing real c_n, we have $c_n = \frac{1}{\sqrt{n}}$, and

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle. \tag{3.8.16}$$

Similarly,

$$|n+1\rangle = d_n \hat{a}^\dagger |n\rangle, \tag{3.8.17}$$

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where d_n is a constant. But

$$1 = \langle n+1|n+1\rangle = |d_n|^2\langle n|\hat{a}\hat{a}^\dagger|n\rangle = |d_n|^2\langle n|[\hat{a},\,\hat{a}^\dagger] + \hat{a}^\dagger\hat{a}|n\rangle = |d_n|^2(1+n). \tag{3.8.18}$$

Therefore, choosing real d_n, we have $d_n = \frac{1}{\sqrt{n+1}}$, and

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \tag{3.8.19}$$

Because the eigenvalue n of $\hat{N}$ satisfies

$$n = \langle n|\hat{N}|n\rangle = |\hat{a}|n\rangle|^2 \geq 0, \tag{3.8.20}$$

it is positive definite or zero. Because $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and we cannot have negative n, we require $\hat{a}|0\rangle = 0$. Therefore, the ground state is $|0\rangle$. The excited states are then constructed by applying the raising operator on $|0\rangle$. In particular,

$$\hat{a}^\dagger|0\rangle = \sqrt{1}|1\rangle \quad \text{or} \quad |1\rangle = \hat{a}^\dagger|0\rangle, \tag{3.8.21}$$

$$\hat{a}^\dagger|1\rangle = \sqrt{2}|2\rangle \quad \text{or} \quad |2\rangle = \frac{1}{\sqrt{2}}\hat{a}^\dagger|1\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2}}|0\rangle, \tag{3.8.22}$$

$$\hat{a}^\dagger|2\rangle = \sqrt{3}|3\rangle \quad \text{or} \quad |3\rangle = \frac{1}{\sqrt{3}}\hat{a}^\dagger|2\rangle = \frac{(\hat{a}^\dagger)^3}{\sqrt{3!}}|0\rangle, \tag{3.8.23}$$

and continuing,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \ldots \tag{3.8.24}$$

The matrix elements of position and momentum follow from (3.8.4), (3.8.16), and (3.8.19),

$$\langle n'|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n'|\left(\hat{a}^\dagger + \hat{a}\right)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{n',n+1} + \sqrt{n}\,\delta_{n',n-1}\right) \tag{3.8.25}$$

and

$$\langle n'|\hat{p}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\langle n'|\left(\hat{a}^\dagger - \hat{a}\right)|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\left(\sqrt{n+1}\,\delta_{n',n+1} - \sqrt{n}\,\delta_{n',n-1}\right). \tag{3.8.26}$$

Note that both $\hat{x}$ and $\hat{p}$ have only off-diagonal matrix elements, which is consistent with $\hat{H}$ or $\hat{N}$ having diagonal elements in the basis with state vectors $|n\rangle$, since $\hat{H}$ does not commute with either one of $\hat{x}$ or $\hat{p}$.

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Question 3.8.2: Why should the ground state be $|0\rangle$ with eigenvalue $n = 0$ for $\hat{N}$? Why not a nonzero constant?

Question 3.8.3: Consider a set of operators $\hat{A}$, $\hat{R}$, and $\hat{L}$ such that $\hat{A}|a\rangle = a|a\rangle$, $[\hat{A}, \hat{R}] = r\hat{R}$, and $[\hat{A}, \hat{L}] = -l\hat{L}$ with $l, r > 0$. Which of the following statements are valid. (A). $|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue a. (B). $\hat{L}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a - l$. (C). $\hat{R}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a + r$. (D). $\hat{L}$ is a lowering operator. (E). $\hat{R}$ is a raising operator.

3.8.3 Wavefunctions in position representation

First, the eigenvalues follow from $\hat{H}|n\rangle = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle$. Therefore,

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \ldots \tag{3.8.27}$$

Next, we want to construct the wavefuctions corresponding to these energy levels in position representation. In particular, the stationary wavefunctions of definite energy are

$$\psi_n(x) = \langle x|n\rangle. \tag{3.8.28}$$

First, let us construct the ground state wavefunction. Using $\hat{a}|0\rangle = 0$,

$$\begin{aligned} 0 &= \hat{a}|0\rangle = \langle x|\hat{a}|0\rangle = \sqrt{\frac{m\omega}{2\hbar}}\langle x|\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)|0\rangle \\ &= \sqrt{\frac{m\omega}{2\hbar}}\left(x\langle x|0\rangle + \frac{i}{m\omega}(-i\hbar\frac{\partial}{\partial x})\langle x|0\rangle\right). \end{aligned} \tag{3.8.29}$$

Thus, we have a differential equation for $\langle x|0\rangle$,

$$\frac{\hbar}{m\omega}\frac{\partial}{\partial x}\langle x|0\rangle + x\langle x|0\rangle = 0. \tag{3.8.30}$$

The solution is

$$\langle x|0\rangle = Ae^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.31}$$

Normalizing,

$$\langle x|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.32}$$

The wavefunctions for the excited states are then obtained by applying the raising operator.

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First let us determine the matrix element of $\langle x'|\hat{a}^\dagger|x''\rangle$,

$$\begin{aligned}
\langle x'|\hat{a}^\dagger|x''\rangle &= \langle x'|\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x}-\frac{i}{m\omega}\hat{p}\right)|x''\rangle = \sqrt{\frac{m\omega}{2\hbar}}\left(\langle x'|\hat{x}|x''\rangle - \frac{i}{m\omega}\langle x'|\hat{p}|x''\rangle\right) \\
&= \sqrt{\frac{m\omega}{2\hbar}}\left(x''\delta(x'-x'') - \frac{i}{m\omega}\left(-i\hbar\frac{\partial\delta(x'-x'')}{\partial x'}\right)\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') - \sigma^2\frac{\partial\delta(x'-x'')}{\partial x'}\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') + \sigma^2\frac{\partial\delta(x'-x'')}{\partial x''}\right) \\
&= \delta(x'-x'')\frac{1}{\sqrt{2}\sigma}\left(x'' - \sigma^2\frac{\partial}{\partial x''}\right), \qquad (3.8.33)
\end{aligned}$$

where we have defined

$$\sigma \equiv \sqrt{\frac{\hbar}{m\omega}} \qquad (3.8.34)$$

for simplicity of notation. Inserting the above expression for the matrix elements and integrating,

$$\begin{aligned}
\langle x|n\rangle &= \langle x|\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle = \frac{1}{\sqrt{n!}}\int dx_1 dx_2 ... dx_{n-1}\langle x|\hat{a}^\dagger|x_1\rangle\langle x_1|\hat{a}^\dagger|x_2\rangle\langle x_2|\cdots|x_{n-1}\rangle\langle x_{n-1}|\hat{a}^\dagger|x_n\rangle\langle x_n|0\rangle \\
&= \frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}\sigma}\left(x-\sigma^2\frac{\partial}{\partial x}\right)\right)^n\langle x|0\rangle \\
&= \frac{1}{\sqrt{n!2^n}\sigma^n}\left(x-\sigma^2\frac{\partial}{\partial x}\right)^n\left(\frac{1}{\pi\sigma^2}\right)^{1/4}e^{-\frac{1}{2\sigma^2}x^2} \qquad (3.8.35) \\
&= \frac{1}{\sqrt{\sqrt{\pi}n!2^n}\,\sigma^{n+\frac{1}{2}}}\left(x-\sigma^2\frac{\partial}{\partial x}\right)^n e^{-\frac{1}{2\sigma^2}x^2}. \qquad (3.8.36)
\end{aligned}$$

If we define

$$\xi = \frac{x}{\sigma}, \qquad (3.8.37)$$

then

$$\langle x|n\rangle = \frac{1}{\sqrt{\sqrt{\pi}n!2^n\sigma}}\left(\xi-\frac{\partial}{\partial\xi}\right)^n e^{-\frac{1}{2}\xi^2} \qquad (3.8.38)$$

This reproduces the wavefunctions obtained after solving the Schrodinger equation. In partic-

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ular, for the ground and first excited states,

$$\langle x|0\rangle = \frac{1}{\sqrt{\sqrt{\pi}\sigma}} e^{-\frac{1}{2}\xi^2} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2} \tag{3.8.39}$$

and

$$\langle x|1\rangle = \frac{1}{\sqrt{\sqrt{\pi}2\sigma}}\left(\xi - \frac{\partial}{\partial\xi}\right) e^{-\frac{1}{2}\xi^2} = \frac{1}{\sqrt{\sqrt{\pi}2\sigma}} 2\xi e^{-\frac{1}{2}\xi^2} = \sqrt{2}\left(\frac{m^3\omega^3}{\pi\hbar^3}\right)^{1/4} x e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.40}$$

3.8.4 Classical simple harmonic oscillator

The probability density for finding the particle at point x in a stationary states $\langle x|n\rangle$ does not evolve in time. In other words, it does not oscillate. Can we construct a quantum state that exhibits a classical-like oscillation?

First let us briefly overview the classical equations of motion of the simple harmonic oscillator in terms of variables that would resemble the quantum oscillator. The Hamiltonian (or total energy) for a classical simple harmonic is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2, \tag{3.8.41}$$

which is the same as (3.8.1) before the position and momentum are promoted to operators. The time evolution of x and p are given by Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \tag{3.8.42}$$

and

$$\dot{p} = -\frac{\partial H}{\partial x} = -m\omega^2 x. \tag{3.8.43}$$

Let us trade x and p for a new complex variable α defined by

$$\sqrt{\hbar\omega}\,\alpha = \sqrt{\frac{m\omega^2}{2}}x + \frac{i}{\sqrt{2m}}p. \tag{3.8.44}$$

With this, we rewrite the Hamiltonian,

$$H = \hbar\omega|\alpha|^2. \tag{3.8.45}$$

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Moreover,

$$x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R, \qquad p = \sqrt{2m\hbar\omega}\,\alpha_I \tag{3.8.46}$$

where α_R and α_I are the real and imaginary parts of α.

The time evolution of α is then

$$\begin{aligned}\sqrt{\hbar\omega}\,\dot{\alpha} &= \sqrt{\frac{m\omega^2}{2}}\dot{x} + \frac{i}{2m}\dot{p} = \sqrt{\frac{m\omega^2}{2}}\frac{p}{m} + \frac{i}{\sqrt{2m}}(-m\omega^2 x)\\ &= \sqrt{\frac{m\omega^2}{2}}\frac{1}{m}\sqrt{2m\hbar\omega}\,\alpha_I - \frac{i}{\sqrt{2m}}m\omega^2\sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R\\ &= -i\sqrt{\hbar\omega^3}\,(\alpha_R + i\alpha_I) = -i\sqrt{\hbar\omega^3}\,\alpha.\end{aligned} \tag{3.8.47}$$

or

$$\dot{\alpha} = -i\omega\alpha. \tag{3.8.48}$$

The solution is

$$\alpha = \alpha_0\, e^{-i\omega t}. \tag{3.8.49}$$

As a specific example for a real α_0, then $\alpha_R = \alpha_0 \cos\omega t$ and $\alpha_I = \alpha_0 \sin\omega t$. In this case, $x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_0 \cos\omega t$ and $p = \sqrt{2m\hbar\omega}\,\alpha_I = \sqrt{2m\hbar\omega}\,\alpha_0 \sin\omega t$. Therefore, the real and imaginary parts of α are related to the position and momentum of the oscillator. In particular, expressing the Hamiltonian of the classical oscillator in terms of α,

$$H = |\alpha|^2\hbar\omega. \tag{3.8.50}$$

3.8.5 The quantum simple harmonic oscillator and coherent state

Question 3.8.4: What is the frequency of oscillation of the probability density for a simple harmonic oscillator in state $|n\rangle$? (A). ω, (B).$\frac{E_n}{\hbar}$, (C).$\frac{E_n - E_{n-1}}{\hbar}$, (D). 0.

Question 3.8.5: What could we do to construct a state whose time evolution could look like a classical oscillator?

Returning back to the quantum oscillator, and noting the expression $H = |\alpha|^2\hbar\omega$ for the classical oscillator, we observe the expectation value of $\hat{H}$,

$$\langle\hat{H}\rangle = \left(\langle\hat{a}^\dagger\hat{a}\rangle + \frac{1}{2}\right)\hbar\omega \tag{3.8.51}$$

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has a similar form as the classical H if we identify $\langle\hat{a}^\dagger\hat{a}\rangle$ with $|\alpha|^2$ and redefine the classical potential energy such that $H \to H = \left(|\alpha|^2 + \frac{1}{2}\right)\hbar\omega$. Thus, we seek a quantum state $|\alpha\rangle$ which gives $\langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle = |\alpha|^2$. This is possible if we could construct a state $|\alpha\rangle$ such that

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{3.8.52}$$

The state $|\alpha\rangle$ that satisfies this equation is called coherent state (or Glauber state, after the recently late Roy Glauber of Harvard). It is a quantum state whose dynamics resembles that of a classical oscillator. We will investigate it in problem set 6.

Question 3.8.6: What is the frequency of oscillation of the probability density of a coherent state of energy $\frac{1}{2}\hbar\omega$? What about for a coherent state of energy $E > \frac{1}{2}\hbar\omega$?

Exercise 3.8.1: Using the raising and lowering operators, answer the following questions for a particle of mass m in the one-dimensional harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2x^2$.

1. The uncertainty in position, Δx, in stationary state $|n\rangle$.
2. The uncertainty in momentum, Δp, in stationary state $|n\rangle$.
3. Check whether the product $\Delta x\,\Delta p$ obeys the uncertainty principle.
4. Rewrite the Hamiltonian $\hat{H}$ in terms of $\hat{\alpha}_R$ and $\hat{\alpha}_I$ defined by $\hat{x} = \sqrt{\frac{2\hbar}{m\omega}}\,\hat{\alpha}_R$ and $\hat{p} = \sqrt{2m\hbar\omega}\,\hat{\alpha}_I$. Rewrite the uncertainty relation in stationary state $|n\rangle$ obtained in part (3) for $\Delta\alpha_R\Delta\alpha_I$.
5. When the oscillator is not in a stationary state, or an eigenstate of the Hamiltonian operator, the uncertainty would take other values than those for individual stationary states obtained above. Show the boundary that separates the region that satisfies the uncertainty principle and the region that does not on a plot of $\Delta\alpha_I$ versus $\Delta\alpha_R$. Indicate the points $(\Delta\alpha_R,\ \Delta\alpha_I)$ on the plane for the first four stationary states corresponding to $n = 0, 1, 2$, and 3.

Exercise 3.8.2: A coherent state (or Glauber state) $|\alpha\rangle$ of the simple harmonic oscillator is an eigenstate of the lowering operator, $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$.

1. Determine the coherent state as a superposition of the stationary states $|n\rangle$ and normalize it.

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2. What is the probability that a system in a coherent state would be found in stationary state $|n\rangle$? What kind of probability distribution function of n is it?

3. Determine the uncertainties Δx, $\Delta\alpha_R$, Δp, and $\Delta\alpha_R$, using the raising and lowering operators, in the coherent state.

4. Show the point for uncertainties $(\Delta\alpha_R, \Delta\alpha_I)$ for the coherent state on the plot you made in part (5). Comparing $\Delta\alpha_R$, $\Delta\alpha_I$, and their product, what is special about the coherent state?

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4 Three-dimensional systems

Now, we begin investigating the Schrödinger equation for a particle in three dimensions with time-independent potential $V(\vec{r})$.

4.1 Three-dimensional infinite square well

Consider a particle of mass m in potential $V(x,y,z) = 0$ for $0 \le x \le l_x$, $0 \le y \le l_y$, $0 \le z \le l_z$, and $V(x,y,z) = \infty$ outside. Because the potential is cubic, the Schrödinger equation in cartesian coordinate is suitable. Therefore, the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi(x,y,z) = E\psi(x,y,z) \tag{4.1.1}$$

for $0 \le x \le l_x$, $0 \le y \le l_y$, and $0 \le z \le l_z$, while $\psi(x,y,z) = 0$ outside. Because there is no potential that mixes the different coordinates, we can use the method of separation of variables, and start with writing,

$$\psi(x,y,t) = \psi_1(x)\psi_2(y)\psi_3(z). \tag{4.1.2}$$

The Schrödinger equation then becomes

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi_1(x) = E_1\psi_1(x), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dy^2}\psi_2(y) = E_2\psi_2(y), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dz^2}\psi_3(z) = E_3\psi_3(z), \tag{4.1.3}$$

where

$$E = E_1 + E_2 + E_3. \tag{4.1.4}$$

We know the solutions from the one-dimensional infinite square well,

$$\psi_{n_x n_y n_z}(x,y,t) = \sqrt{\frac{8}{l_x l_y l_z}}\sin(\frac{n_x\pi}{l_x}x)\sin(\frac{n_y\pi}{l_y}z)\sin(\frac{n_z\pi}{l_z}z) \tag{4.1.5}$$

and

$$E_{n_x n_y n_z} = \frac{\hbar^2\pi^2}{2m}\left(\frac{n_x^2}{l_x^2}+\frac{n_y^2}{l_y^2}+\frac{n_z^2}{l_z^2}\right) \tag{4.1.6}$$

with $n_x, n_y, n_z = 1, 2, \ldots$ Note that the wave vector is

$$\vec{k} = (k_x, k_y, k_z) = (\frac{n_x\pi}{l_x}, \frac{n_y\pi}{l_y}, \frac{n_z\pi}{l_z}). \tag{4.1.7}$$

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Exercise 4.1.1: When we talked about Planck's black-body radiation formula, we mentioned that the number density (per unit volume per unit frequency) of states in a cavity was $\frac{\mathcal{D}(\omega)}{V} = \frac{\omega^2}{\pi^2 c^3}$. Show that considering the standing modes of the three-dimensional infinite square well in a large cubic box.

Solution: Considering a large box, we can view its volume in $\vec{k}$–space as spherical, and the volume of spherical shell of infinitesimal thickness dk at radius k is $\frac{1}{8} 4\pi k^2 dk = \frac{1}{2}\pi k^2 dk$, where the factor of $\frac{1}{8}$ because all components of $\vec{k}$ are positive. The volume of a single cell is $\vec{k}$–space is $\frac{\pi}{l_x}\frac{\pi}{l_y}\frac{\pi}{l_z} = \frac{\pi^2}{V}$, where $V = l_x l_y l_z$ is the volume of the box. The number of states within this volume is $dN = 2\frac{\frac{1}{2}\pi k^2 dk}{\frac{\pi^3}{V}} = \frac{V}{\pi^2} k^2 dk$, where the factor of 2 is because radiation has two possible polarizations, or a standing plane wave has two independent directions (for describing the plane on which the electric field lives). Considering radiation, $k = \frac{\omega}{c}$. Thus, $dN = 2\frac{V}{\pi^2}\frac{\omega^2}{c^2}\frac{d\omega}{c}$ and $\frac{\mathcal{D}(\omega)}{V} = \frac{dN}{d\omega V} = \frac{\omega^2}{\pi^2 c^3}$.

Exercise 4.1.2: The Hamiltonian operator for a particle of mass m constrained to undergo circular motion on a frictionless track is given by $\hat{H} = -\frac{\hbar^2}{2mR^2}\frac{\partial^2}{\partial\phi^2}$, where R is the radius of the circle and ϕ is the angular coordinate. What boundary condition should the wavefunction $\psi(\phi)$ satisfy? Using the Schrödinger equation and the boundary condition, determine the wavefunctions and energies of the stationary states. Determine the degeneracy of each energy level.

Exercise 4.1.3: Consider the three-dimensional harmonic oscillator with potential $V(x, y, z) = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$. Answer the following questions using cartesian coordinates, separation of variables, and your knowledge of the energies and wavefunctions of the one-dimensional simple harmonic oscillator. Write down the energy levels in terms of three quantum numbers n_1, n_2, and n_3. Write the same energy levels in terms of one quantum number n. Show that the nth energy level is $\frac{1}{2}(n+1)(n+2)$–fold degenerate.

4.2 The Schrödinger equation in spherical coordinates

The Schrödinger equation in three dimensions in position representation in spherical coordinates is

$$i\hbar\frac{\partial\psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t) + V(\vec{r})\psi(\vec{r}, t), \tag{4.2.1}$$

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where $\vec{r} = (r, \theta, \phi)$, $\psi(\vec{r}, t) = \psi(r, \theta, \phi, t)$ and

$$\vec{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \tag{4.2.2}$$

is the Laplacian operator.

For a time-independent and central potential, $V(\vec{r}) = V(r)$. In other words, the potential depends only of the radial coordinate r but not on θ and ϕ. The time-independent Schrödinger equation then becomes

$$-\frac{\hbar^2}{2mr^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,\theta,\phi)}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi(r,\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi(r,\theta,\phi)}{\partial\phi^2}\right) + V(r)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi). \tag{4.2.3}$$

Because the potential does not mix the different coordinates, we can make separation of variables and write

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi). \tag{4.2.4}$$

The Schrödinger equation then reduces to

$$\frac{1}{R(r)}\left[\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r) - E)\right]R(r) = -\frac{1}{Y(\theta,\phi)}\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dY(\theta,\phi)}{d\theta}\right) + \frac{1}{\sin^2\theta}\frac{d^2Y(\theta,\phi)}{d\phi^2}\right]. \tag{4.2.5}$$

Each side must be a constant and equal. Thus, denoting the constant by $l(l+1)$ for convenience in notation later,

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y(\theta,\phi)}{\partial\phi^2} = -l(l+1)Y(\theta,\phi). \tag{4.2.6}$$

and

$$\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r) - E)R(r) = l(l+1)R(r). \tag{4.2.7}$$

4.3 Orbital angular momentum

The orbital angular momentum in classical mechanics is defined by $\vec{L} = \vec{r}\times\vec{p}$. In quantum mechanics, this is promoted to an operator, and in position representation,

$$\hat{\vec{L}} = \hat{\vec{r}}\times\hat{\vec{p}}. \tag{4.3.1}$$

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In cartesian coordinates, the components of the angular momentum operator are

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \tag{4.3.2}$$

where we need to distinguish the hat over an operator versus over a unit vector from context. Because the components of the position and the momentum operators do not commute, the components of the angular momentum operator do not commute either. In particular,

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] = \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x] + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\ &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y = i\hbar\hat{L}_z \end{aligned} \tag{4.3.3}$$

Similarly, and collecting,

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_z, \quad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y, \tag{4.3.4}$$

or

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \tag{4.3.5}$$

with $i = 1, 2, 3$ representing the x, y, and z components respectively, and ϵ_{ijk} is the epsilon tensor with components defined by

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1, \tag{4.3.6}$$

it becomes -1 for odd permutation of indices, such as $\epsilon_{213} = -1$, and vanishes when any index is repeated, such as $\epsilon_{112} = 0$.

Furthermore, the square of the total angular momentum operator is

$$\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \tag{4.3.7}$$

and it commutes with each component of the total angular momentum. For instance,

$$\begin{aligned} [\hat{\vec{L}}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] = \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\ &= -i\hbar\hat{L}_x\hat{L}_y - i\hbar\hat{L}_y\hat{L}_x + i\hbar\hat{L}_y\hat{L}_x, + i\hbar\hat{L}_x\hat{L}_y = 0. \end{aligned} \tag{4.3.8}$$

Thus,

$$[\hat{\vec{L}}^2, \hat{L}_i] = 0. \tag{4.3.9}$$

Question 4.3.1: Out of the four operators, $\hat{\vec{L}}$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could

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we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

Question 4.3.2: Out of the four operators, $\hat{\vec{L}}^2$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

In position representation,

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = -i\hbar \vec{r} \times \vec{\nabla} \tag{4.3.10}$$

In spherical coordinates,

$$\begin{aligned} \hat{\vec{L}} &= -i\hbar r\hat{r} \times \left(\hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} \right) \\ &= -i\hbar \left(\hat{\phi}\frac{\partial}{\partial \theta} - \hat{\theta}\frac{1}{\sin\theta}\frac{\partial}{\partial \phi} \right). \end{aligned} \tag{4.3.11}$$

The components along cartesian unit vectors can be obtained using the unit vector relations,

$$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z} \tag{4.3.12}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z} \tag{4.3.13}$$

$$\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}. \tag{4.3.14}$$

With these,

$$\begin{aligned} \hat{\vec{L}} &= -i\hbar \left((-\sin\phi\,\hat{x} + \cos\phi\,\hat{y})\frac{\partial}{\partial \theta} - (\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z})\frac{1}{\sin\theta}\frac{\partial}{\partial \phi} \right) \\ &= L_x\hat{x} + L_y\hat{y} + L_z\hat{z}, \end{aligned} \tag{4.3.15}$$

where

$$\hat{L}_x = i\hbar \left(\sin\phi\,\frac{\partial}{\partial \theta} + \cot\theta\cos\phi\,\frac{\partial}{\partial \phi} \right) \tag{4.3.16}$$

$$\hat{L}_y = i\hbar \left(-\cos\phi\,\frac{\partial}{\partial \theta} + \cot\theta\sin\phi\,\frac{\partial}{\partial \phi} \right) \tag{4.3.17}$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial \phi} \tag{4.3.18}$$

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Squaring, $\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$,

$$\hat{\vec{L}}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right] \tag{4.3.19}$$

With this, the angular part of the Schrödinger equation given by (4.2.6) can be rewritten as

$$\hat{\vec{L}}^2 Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi). \tag{4.3.20}$$

We also rearrange and rewrite the radial part given by (4.2.7) as

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) + \left(V(r) + \frac{l(l+1)\hbar^2}{2mr^2}\right)R(r) = ER(r). \tag{4.3.21}$$

Or, rewriting again,

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - V_{\text{eff}}(r)R(r) = ER(r), \tag{4.3.22}$$

which is a Schrödinger like equation for the radial part of the wavefunction with effective potential,

$$V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}. \tag{4.3.23}$$

Note that the second term is the potential corresponding to centrifugal force in classical mechanics due to the orbital motion of the particle.

4.3.1 Spherical harmonics

We want to find the solutions to the angular equation $\hat{\vec{L}}^2 Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi)$; i.e.,

$$-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi). \tag{4.3.24}$$

Using separation of variables,

$$Y(\theta,\phi) = \Theta(\theta)\Phi(\phi) \tag{4.3.25}$$

we obtain

$$\frac{1}{\Theta}\left[\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + l(l+1)\sin^2\theta\right] = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = constant = m^2. \tag{4.3.26}$$

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The solution for the ϕ equation has the simple form

$$\Phi(\phi) = e^{im\phi}. \tag{4.3.27}$$

Demanding that $\Phi(\phi)$ be periodic in ϕ with period 2π gives the constraint

$$m = 0, \pm 1, \pm 2, \cdots . \tag{4.3.28}$$

The equation for $\Theta(\theta)$ can be written in terms of a new variable defined by

$$x \equiv \cos\theta \tag{4.3.29}$$

as

$$(1-x^2)\frac{d^2P(x)}{dx^2} - 2x\frac{dP(x)}{dx} + \left(l(l+1) - \frac{m^2}{1-x^2}\right)P(x) = 0. \tag{4.3.30}$$

This is the associated Legendre differential equation whose solutions can be obtained using the method of power series. For the special case of $m = 0$, the equation is called the Legendre differential equation, and the solutions are called the Legendre polynomials.

The solutions to the associated Legendre differential equation are called the associated Legendre functions,

$$P_l^{m_l}(x) = (1-x^2)^{|m_l|/2}\left(\frac{d}{dx}\right)^{|m_l|}P_l(x), \tag{4.3.31}$$

where $P_l(x)$ is the lth Legendre polynomial given by the Rodrigues formula

$$P_l(x) = \frac{1}{2^l l!}\left(\frac{d}{dx}\right)^l (x^2-1)^l \tag{4.3.32}$$

with l taking values

$$l = 0, 1, 2, ... \tag{4.3.33}$$

and for each l, m_l takes $2l+1$ values

$$m_l = -l, -l+1, ..., l-1, l. \tag{4.3.34}$$

The normalized solutions to the angular part, including both θ and ϕ, are the spherical harmonics,

$$Y_l^{m_l}(\theta,\phi) = (-1)^{m_l}\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-|m_l|)!}{(l+|m_l|)!}}e^{im_l\phi}P_l^{m_l}(\cos\theta). \tag{4.3.35}$$

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for $m_l \geq 0$, and

$$Y_l^{m_l}(\theta,\phi) = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-|m_l|)!}{(l+|m_l|)!}}e^{im_l\phi}P_l^{m_l}(\cos\theta). \tag{4.3.36}$$

for $m_l < 0$. $Y_l^{m_l}(\theta,\phi)$ are simultaneous eigenfunctions of $\hat{\vec{L}}^2$ and $\hat{L}_z$ with

$$\hat{\vec{L}}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}, \qquad \hat{L}_z Y_l^{m_l} = m_l\hbar Y_l^{m_l}. \tag{4.3.37}$$

The first few Legendre polynomials are listed below.

$$\begin{aligned} P_0(x) &= 1 & P_0(\theta) &= 1 \\ P_1(x) &= x & P_1(\theta) &= \cos\theta \\ P_2(x) &= \frac{1}{2}(3x^2-1) & P_2(\theta) &= \frac{1}{2}(3\cos^2\theta - 1) \end{aligned} \tag{4.3.38}$$

The Legendre polynomials are normalized such that they satisfy the orthogonality relation

$$\int_{-1}^{1} P_{l'}(x)P_l(x)\,dx = \int_0^{\pi} P_{l'}(\theta)P_l(\theta)\sin\theta d\theta = \frac{2}{2l+1}\delta_{l'l} \tag{4.3.39}$$

The first few associated Legendre functions are listed below.

$$\begin{aligned} P_0^0(x) &= 1 & P_0^0(\theta) &= 1 \\ P_1^1(x) &= \sqrt{1-x^2} & P_1^1(\theta) &= \sin\theta \\ P_1^0(x) &= x & P_1^0(\theta) &= \cos\theta \\ P_2^2(x) &= 3(1-x^2) & P_2^2(\theta) &= 3\sin^2\theta \\ P_2^1(x) &= 3x\sqrt{1-x^2} & P_2^1(\theta) &= 3\cos\theta\sin\theta \\ P_2^0(x) &= \frac{1}{2}\left(3x^2-1\right) & P_2^0(\theta) &= \frac{1}{2}\left(3\cos^2\theta - 1\right) \end{aligned} \tag{4.3.40}$$

with

$$P_l^{-m_l}(x) = P_l^{m_l}(x). \tag{4.3.41}$$

The associated Legendre functions are normalized such that they satisfy the orthogonality relation

$$\int_{-1}^{1} P_{l'}^{m_l'}(x)P_l^{m_l}(x)\,dx = \int_0^{\pi} P_{l'}^{m_l'}(\theta)P_l^{m_l}(\theta)\sin\theta d\theta = \frac{(l+m_l)!}{(2l+1)(l-m_l)!}\delta_{l'l}\delta_{m_l'm_l} \tag{4.3.42}$$

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The first few spherical harmonics are given by

$$\begin{aligned} Y_0^0(\theta,\phi) &= \frac{1}{\sqrt{4\pi}} \\ Y_1^{\pm 1}(\theta,\phi) &= \mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\phi} \\ Y_1^0(\theta,\phi) &= \sqrt{\frac{3}{4\pi}}\cos\theta \\ Y_2^{\pm 2}(\theta,\phi) &= \sqrt{\frac{15}{32\pi}}\sin^2\theta\, e^{\pm 2i\phi} \qquad (4.3.43) \\ Y_2^{\pm 1}(\theta,\phi) &= \mp\sqrt{\frac{15}{8\pi}}\sin\theta\; e^{\pm i\phi} \qquad (4.3.44) \\ Y_2^0(\theta,\phi) &= \sqrt{\frac{5}{16\pi}}\left(3\cos^2\theta - 1\right) \qquad (4.3.45) \end{aligned}$$

The spherical harmonics satisfy the orthogonality relation

$$\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta\, Y_{l'}^{m_l'*}(\theta,\phi) Y_l^{m_l}(\theta,\phi) = \delta_{l'l}\delta_{m_l' m_l}. \qquad (4.3.46)$$

The eigenvalue problem of orbital angular momentum $\hat{\vec{L}}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}$ and $\hat{L}_z Y_l^{m_l} = m\hbar Y_l^{m_l}$ was solved above in position representation. The quantum numbers, or eigenvalues for the simultaneously diagonalizable operators $\hat{\vec{L}}^2$ and $\hat{L}_z$, come in a pair of numbers l, m_l. We can write this eigenstate as a ket in abstract vector space as $|lm_l\rangle$, and we have

$$\hat{\vec{L}}^2 |lm_l\rangle = l(l+1)\hbar^2 |lm_l\rangle, \qquad \hat{L}_z |lm_l\rangle = m\hbar |lm_l\rangle. \qquad (4.3.47)$$

The spherical harmonics are then the wavefunctions in position representation

$$Y_l^{m_l}(\theta,\phi) = \langle \vec{r} | lm_l \rangle. \qquad (4.3.48)$$

Exercise 4.3.1:

Consider a "rigid rotor" model for the rotational motion of H_2 molecule with two H atoms, each of mass M, at fixed separation of $2R$.

1. Considering a molecule rotating about an axis perpendicular to the center of the interatomic separation, write down the classical Hamiltonian for the rotational energy of the rotor in terms of angular momentum $\vec{L}$ and moment

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of inertia $I = 2MR^2$.

2. Promote the classical Hamiltonian to a Hamiltonian operator, and write down the wavefunctions in spherical coordinates and the corresponding energies.

3. Using the mass of the proton for the mass of the hydrogen atom and taking $2R = 0.15\,nm$, determine the numerical value of the first four energy levels in eV.

4.3.2 Parity of the spherical harmonics

The parity operator takes $\vec{r}$ to $-\vec{r}$. In cartesian coordinates

$$\hat{P}\psi(x, y, z) = \psi(-x, -y, -z). \tag{4.3.49}$$

In spherical coordinates,

$$\hat{P}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi). \tag{4.3.50}$$

In particular, for the Legendre polynomials, noting that

$$P_l^{m_l}(\theta) = (1 - \cos^2\theta)^{|m_l|/2}\left(\frac{d}{d\cos\theta}\right)^{|m_l|}\frac{1}{2^l l!}\left(\frac{d}{d\cos\theta}\right)^l(\cos^2\theta - 1)^l \tag{4.3.51}$$

and $\hat{P}\cos\theta = \cos(\pi - \theta) = -\cos\theta$, we have

$$\hat{P}P_l^{m_l}(\theta) = P_l^{m_l}(\pi - \theta) = (-1)^{l+|m_l|}P_l^{m_l}(\theta) = (-1)^{l-|m_l|}P_l^{m_l}(\theta) \tag{4.3.52}$$

Therefore, the Legendre polynomials are even for $l + |m_l|$ even and odd for $l + |m_l|$ odd. Furthermore, for the azimuthal part of the wavefunction,

$$\hat{P}e^{im_l\phi} = e^{im_l(\phi+\pi)} = (-1)^{m_l}e^{im_l\phi}. \tag{4.3.53}$$

The spherical harmonics are products of the two, and

$$\hat{P}Y_l^{m_l}(\theta, \phi) = Y_l^{m_l}(\pi - \theta, \phi + \pi) = (-1)^{l-|m_l|+m_l}Y_l^{m_l}(\theta, \phi) = (-1)^l Y_l^{m_l}(\theta, \phi). \tag{4.3.54}$$

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4.4 The hydrogen atom

Question 4.4.1: What is the largest set of simultaneously diagonalizable operators among the position, linear momentum, and angular momentum operators in addition to the Hamiltonian for a system with central potential? (A). $\hat{H}$, $\hat{\vec{r}}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$ (B). $\hat{H}$, $\hat{\vec{r}}$, $\hat{\vec{L}}$ (C). $\hat{H}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$ (D). $\hat{H}$, $\hat{\vec{L}}^2$, $\hat{\vec{L}}_z$ (E). $\hat{H}$, $\hat{\vec{L}}^2$, $\hat{\vec{L}}_x$.

Exercise 4.4.1: Show that

1. $[\hat{\vec{L}}_z, \hat{\vec{p}}^2] = 0$,
2. $[\hat{\vec{L}}_z, \hat{V}(r)] = 0$

Answer: We could see that $[\hat{\vec{L}}_z, \hat{\vec{p}}^2] = 0$ without doing the math, since $\hat{\vec{p}}^2$ represents the magnitude (or length) of linear momentum squared, which is invariant under rotation. Moreover, $[\hat{\vec{L}}_z, \hat{V}(r)] = 0$, since a central potential involves only the length of the position vector, $r = |\vec{r}|$, which is invariant under rotation. Alternatively, we could do the math as follows.

1.

$$\begin{aligned}[\hat{\vec{L}}_z, \hat{\vec{p}}^2] &= [\hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2] = [\hat{x}, \hat{p}_x^2]\hat{p}_y - [\hat{y}, \hat{p}_y^2]\hat{p}_x \\ &= 2i\hbar\hat{p}_x\hat{p}_y - 2i\hbar\hat{p}_y\hat{p}_x = 2i\hbar[\hat{p}_x, \hat{p}_y] = 0. \end{aligned} \tag{4.4.1}$$

2.

$$[\hat{\vec{L}}_z, \hat{V}(r)] = [-i\hbar\frac{\partial}{\partial\phi}, \hat{V}(r)] = 0, \tag{4.4.2}$$

since ϕ and r independent spherical coordinates.

Question 4.4.2: Which of the following are true about a system with central potential? (A). Total angular momentum is conserved. (B). Each component of angular momentum is conserved. (C). Total linear momentum is conserved. (D). Each component of linear momentum is conserved.

For the specific case of the hydrogen atom, which contains one proton in the nucleus and one electron at distance r from it, the potential in position representation is

$$\hat{V} = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}. \tag{4.4.3}$$

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The radial part of the Schrödinger equation given by (4.3.21)

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right)+\left(-\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}+\frac{l(l+1)\hbar^2}{2mr^2}\right)R(r)=ER(r). \tag{4.4.4}$$

Note that the radial equation is a Schrödinger like equation with effective potential

$$V_{\text{eff}}(r)=V(r)+\frac{l(l+1)\hbar^2}{2mr^2} \tag{4.4.5}$$

with $V(r)=-\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$. The second term, $\frac{l(l+1)\hbar^2}{2mr^2}$ arises as a centrifugal term because of angular momentum for $l\neq 0$. How the effective potential comes about as a sum of the Coulomb potential and the cetrifugal term are shown in the figure below.

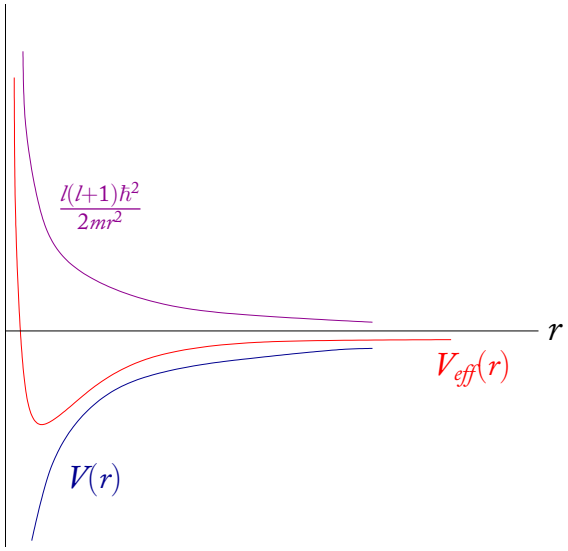


Figure 4.4.1: Effective potential for hydrogen atom.

Exercise 4.4.2:

1. Show that the radial part of the Schödinger equation for the electron of mass m in a hydrogen atom with potential $V=-\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$; i.e., $-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2}+\left(-\frac{e^2}{4\pi\epsilon_0}\frac{1}{r}+\frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right)u=Eu$, where $u(r)=rR(r)$, can be written as $\frac{d^2u}{d\rho^2}=\left(1-\frac{\rho_0}{\rho}+\frac{l(l+1)}{\rho^2}\right)u$, where $\rho=\kappa r$, $\rho_0=\frac{me^2}{2\pi\epsilon_0\hbar^2\kappa}$, $\kappa=\sqrt{-\frac{2mE}{\hbar^2}}$, and l is the total angular quantum number.

2. Investigate the asymptotic large ρ behavior of the radial equation and show that normalizable solutions could be written as $u(\rho)=\rho^{l+1}e^{-\rho}v(\rho)$ with the

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equation $\rho\frac{d^2v}{d\rho^2} + 2(l+1-\rho)\frac{dv}{d\rho} + (\rho_0 - 2(l+1))v = 0$ for $v(\rho)$.

3. Show that the method of power series with $v(\rho) = \sum_{j=0}^{\infty} c_j\rho^j$ produces the recursion relations $c_{j+1} = \frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)}c_j$.

4. Using the large j behavior of the recursion relation, argue how the infinite series would give $v(\rho) \sim e^{2\rho}$ and reintroduce a divergent solution.

5. Truncating the series, show that the energy levels are quantized and given by $E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2 n^2} = -\frac{13.6\,eV}{n^2}$, where $n = 1, 2, 3, \cdots$. Show that the nth energy level is $2n^2-$fold degenerate.

6. Make a plot of the four lowest energy levels to scale, and indicate the list of associated quantum numbers (nlm_l) for each energy level.

7. Assuming that a hydrogen atom making a transition from a higher energy level with principal quantum number m to a lower energy level n gives off all its energy by releasing a photon, write down the expression for the frequency of the emitted radiation. Show that the possible frequencies can be written as $f = R_\infty\, c\,(\frac{1}{n^2} - \frac{1}{m^2})$, where $R_\infty \approx 1.1 \times 10^7 m^{-1}$ is the Rydberg constant and c is the speed of light.

The requirement of normalizable wavefunctions leads to quantized energy levels and the corresponding wavefunctions. As we see it in the problem set, the energies are given by

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2 n^2} = -\frac{13.6\,eV}{n^2}; \quad n = 1, 2, 3, \cdots \tag{4.4.6}$$

The energy levels can be rewritten in terms of the fine structure constant defined by

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \tag{4.4.7}$$

and the Bohr radius given by

$$a = \frac{4\pi\epsilon_0\hbar^2}{me^2}, \tag{4.4.8}$$

as

$$E_n = -\frac{1}{2}\,\alpha^2\, mc^2\frac{1}{n^2}. \tag{4.4.9}$$

or

$$E_n = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{2a}\frac{1}{n^2}.. \tag{4.4.10}$$

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The wavefunctions are

$$\psi_{nlm_l}(r,\theta,\phi) = R_{nl}(r)Y_l^{m_l}(\theta,\phi), \tag{4.4.11}$$

with

$$R_{nl} = \frac{1}{r}\rho^{l+1}e^{-\rho}v(\rho), \tag{4.4.12}$$

where

$$v(\rho) = \sum_{j=0}^{j_{\max}} c_j\rho^j, \tag{4.4.13}$$

$$c_{j+1} = \frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)}c_j, \tag{4.4.14}$$

and $j_{\max}$ is the order of the polynomial at which we truncate the series such that

$$2(j_{\max}+l+1)-\rho_0 = 0, \tag{4.4.15}$$

$$\rho_0 = \frac{me^2}{2\pi\epsilon_0\hbar^2}\sqrt{-\frac{\hbar^2}{2mE}} \tag{4.4.16}$$

$$j_{\max} = 0,1,2,... \tag{4.4.17}$$

We define a new integer

$$n = j_{\max}+l+1 = \frac{\rho_0}{2}, \tag{4.4.18}$$

and

$$n = 1,2,3,... \tag{4.4.19}$$

is called principal quantum number, which with (4.4.16) gives E_n. Furthermore, the angular momentum quantum number l for a fixed n takes possible values

$$l = 0,1,2,...,n-1. \tag{4.4.20}$$

The quantum number m_l for the $z-$component of angular momentum takes $2l+1$ possible values

$$m_l = -l,-m_l+1,...,l \tag{4.4.21}$$

The degeneracy of the nth energy level is

$$g(n) = \sum_{l=0}^{n-1} 2(2l+1) = 2n^2, \tag{4.4.22}$$

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where the factor of 2 is because of two possible spin orientations (spin-up or spin-down) of the electron. The radial part of the wavefunction can be written as

$$v(\rho) = L^{2l+1}_{n-l-1}(2\rho), \tag{4.4.23}$$

where

$$L^p_{q-p}(x) \equiv (-1)^p \left(\frac{d}{dx}\right)^p L_q(x) \tag{4.4.24}$$

is called associated Laguerre polynomial with

$$L_q(x) \equiv e^x \frac{1}{q!}\left(\frac{d}{dx}\right)^q (e^{-x}x^q) \tag{4.4.25}$$

called the Laguerre polynomial.

The first few Laguerre polynomials are

$$\begin{aligned} L_0(x) &= 1 \\ L_1(x) &= -x+1 \\ L_2(x) &= \frac{1}{2}x^2 - 2x + 1. \end{aligned} \tag{4.4.26}$$

The first few associated Laguerre polynomials are

$$\begin{aligned} L^0_0(x) &= 1 & L^2_0(x) &= 1 \\ L^0_1(x) &= -x+1 & L^1_1(x) &= -x+2 \\ L^1_0(x) &= 1 & L^2_1(x) &= -x+3 \\ L^0_2(x) &= \frac{1}{2}x^2 - 2x + 1 & L^2_2(x) &= \frac{1}{2}x^2 - 4x + 6 \end{aligned} \tag{4.4.27}$$

The first few radial wavefunctions are The first few Laguerre polynomials are

$$\begin{aligned} R_{10}(r) &= \frac{2}{\sqrt{a^3}} e^{-\frac{r}{a}} \\ R_{20}(r) &= \frac{1}{\sqrt{2a^3}}\left(1 - \frac{1}{2}\frac{r}{a}\right) e^{-\frac{r}{2a}} \\ R_{21}(r) &= \frac{1}{\sqrt{24a^3}}\frac{r}{a} e^{-\frac{r}{2a}}, \end{aligned} \tag{4.4.28}$$

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where

$$a = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \times 10^{-10}\, m \tag{4.4.29}$$

is the Bohr radius.

The ground state wavefunction is

$$\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi a^3}} e^{-\frac{r}{a}} \tag{4.4.30}$$

Putting the radial and the angular parts together, the normalized wavefunctions of the hydrogen atom are

$$\psi_{nlm_l}(r,\theta,\phi) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-\frac{r}{na}} \left(\frac{2r}{na}\right)^l \left[L^{2l+1}_{n-l-1}\left(\frac{2r}{na}\right)\right] Y_l^{m_l}(\theta,\phi). \tag{4.4.31}$$

These satisfy the orthonormality relation

$$\int \psi^*_{n'l'm'_l} \psi_{nlm_l} r^2 \sin\theta dr d\theta d\phi = \delta_{n'n}\delta_{l'l}\delta_{m'_l m_l}. \tag{4.4.32}$$

Demo 4.4.1: Hydrogen spectrum. We want to investigate the numerical values of radiation emitted in atomic transitions of hydrogen and find out which ones are in the visible range compared to what you see through the diffraction grating, and check if it matches what we see through a diffraction grating. We also view the spectra from He, Ne and Ar atoms in addition to that from an incandescent bulb.

Focusing on H atom, transitions that end in the ground state are called the Lyman series. Those that end in the first excited state are called the Balmer series. Those that end in the second excited state are called the Paschen series. These transitions are shown in the figure below.

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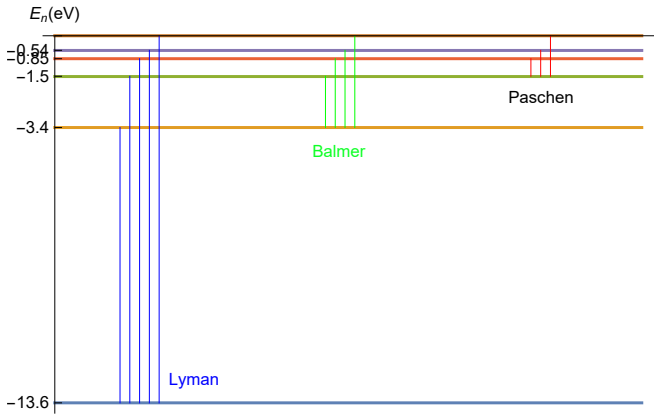


Figure 4.4.2: Hydrogen spectrum.

First, let us recall that red comes in $620-670\,nm$, blue in $450-495\,nm$, green in $495-570\,nm$, and violet in $380-450\,nm$. The transition from $m=2$ to $n=1$ gives $\Delta E = E_2 - E_1 = -3.4eV + 13.6eV = 10.2eV$, which corresponds to $\lambda = \frac{hc}{\Delta E} \approx 120nm$. This is the ultraviolet but not in the visible range. Because all other Lyman transitions have higher frequency and lower wavelength, none of them are in the visible range. Thus, we need to move to the Balmer series. The spectral lines we see using the diffraction grating correspond to individual transitions. We have diffraction gratings with 500 slits per millimeter. The angular separation for constructive interference patterns from a grating of separation d between slits is given

$$\sin\theta = \frac{\lambda}{d}, \tag{4.4.33}$$

where $d = \frac{1mm}{500} = 0.002mm$. We have the following table for the wavelengths in the Balmer series and the angles seen through the grating. We ignore the higher frequencies (shorter wavelengths), since they get too crowded together as the energy levels get very close, making it difficult to see them as separate spectral lines. These look like a single line.

ΔE	$\lambda = \frac{hc}{\Delta E}$	Color	Angle
$E_3 - E_2 = 1.89\,eV$	$656\,nm$	red	18.8^0
$E_4 - E_2 = 2.55\,eV$	$486\,nm$	cyan (blue-green)	13.9^0
$E_5 - E_2 = 2.86\,eV$	$434\,nm$	violet	12.4^0
$E_6 - E_2 = 3.02\,eV$	$410\,nm$	violet	11.8^0

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It is important to note the differences in the spectra from the incandescent bulb and H, He, Ne, and Ar atoms. The emission from the incandescent bulb is due to thermal radiation arising from the jiggling motion of charges, and it contains a broad spectrum of frequencies. On the other hand, the spectrum from the atoms arises from transitions between discrete energy levels, resulting in spectral lines. Note the red color in H atom has a specific location that is different from a red in another atom, such as He. This has applications in various fields such as observational astronomy, where the spectroscopy of radiation could be used to find out the atomic composition of the surface of a star.

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